

# WARDA SHAHZAD

PhD Chemistry • Kesterite (CZTS) Thin-Film Photovoltaics • Defect Engineering & Optical Characterization

## CONTACT

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## RESEARCH IMPACT

**3** first-author papers

## KEY EXPERTISE

- CZTS kesterite thin-film absorbers, single- and co-doped alkali (Na, Li, K) systems
- Solution-based chemical deposition synthesis
- Full XRD structural analysis: lattice constants, crystallite size, dislocation density, microstrain, Williamson-Hall modeling
- SEM/EDX, UV-Vis spectroscopy
- Novel protocol design (two-step annealing, coprecipitation synthesis)
- Photophysical modeling: Tauc plots, single oscillator model, dielectric constant, optical/electrical/thermal conductivity
- VESTA crystal structure visualization

## LANGUAGES

Urdu — Native  
English — Fluent

## REFERENCES

(Dept. of Chemistry, COMSATS University Islamabad, Abbottabad Campus, Pakistan)

**Prof. Dr. Bushra Ismail**

PhD Supervisor —  
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**Prof. Dr. Rafaqat A. Khan**

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## RESEARCH PROFILE

I'm a PhD researcher in Materials Chemistry building a focused research line in  $\text{Cu}_2\text{ZnSnS}_4$  (CZTS) kesterite thin-film solar absorbers. My work centers on solution-based synthesis and alkali-cation (Na, Li, K) doping to engineer defects, suppress secondary phases, and tune optical bandgaps, the core levers the field uses to push kesterite devices toward higher efficiency. I translate tight composition, processing, and property narratives into first-author publications in ranked international journals, and I'm now looking for a postdoctoral position to extend this expertise to new synthesis platforms and semiconductor material systems.

## EDUCATION & TRAINING

**PhD, Chemistry** — COMSATS University Islamabad, Abbottabad Campus

Fall 2020 – Spring 2026 | Thesis submitted for foreign evaluation, external viva pending

- Thesis: “Exploration of the Physical Properties of Kesterite Thin Films Modified by Alkali Metal Cations for Sunlight Harvesting,” extended to Li/Na and Li/K co-doped CZTS systems (manuscript in preparation).
- Milestones: comprehensive exam (Fall 2021), synopsis defense (Fall 2022), pre-DAC (Fall 2025), DAC and internal seminar (Spring 2026).

Supervisor: Bushra Ismail

**MS, Chemistry** — COMSATS University Islamabad, Abbottabad Campus

Fall 2017 – Spring 2019

- Thesis on visible-light photocatalysis using zirconate perovskite/graphene oxide composites. Developed a novel coprecipitation synthesis route for Sr/Ba/Mn-zirconate perovskites in place of the conventional hydrothermal method, published in *Ceramics International* (2022).

Supervisor: Bushra Ismail

**MSc, Chemistry** — University of Wah, Pakistan

Fall 2015 – Spring 2017

- Research internship, Pakistan Ordnance Factories Institute of Technology, Wah Cantt (Jul–Aug 2017).

## RESEARCH EXPERIENCE

**Doctoral Researcher, CZTS Thin-Film Solar Absorbers**

COMSATS University Islamabad, Abbottabad Campus | 2020 – 2026

- Built a reproducible structure-property pattern across Li-, Na-, and K-doped CZTS: alkali incorporation consistently narrows the bandgap (0.06–0.5 eV, depending on dopant) while cutting dislocation density 2–4×.
- Na-doped CZTS: narrowed the bandgap from 1.5 to 1.0 eV, reduced microstrain from  $0.37 \times 10^{-3}$  to  $0.13 \times 10^{-3}$ , first direct structural-optical link for Na-driven antisite suppression.
- Li-doped CZTS: designed and validated a two-step annealing protocol, among the first applied to this system, tuning the bandgap from 1.36 to 1.27 eV and cutting dislocation density from 7.3 to 2.13 ( $\times 10^{14} \text{ m}^{-2}$ ).
- K-doped CZTS (under review): grew crystallite size from 40 to 52 nm and tuned the bandgap from 1.50 to 1.42 eV.
- Extended this framework to Li/Na and Li/K co-doped CZTS systems to probe synergistic defect suppression; included in doctoral thesis, manuscript in preparation.
- Independently built crystal structure models in VESTA and ran the full XRD, SEM/EDX, and UV-Vis workflow from data collection through interpretation.
- Mentored MS students in materials chemistry synthesis and characterization within the lab.

## PUBLICATIONS & RESEARCH OUTPUT

**First-Author, PhD Research (2 published, 1 under review)**

- [Journal of Physics and Chemistry of Solids \(2025\)](#): Na-doped CZTS. Bandgap narrowed 1.5→1.0 eV; first direct link between Na-driven antisite suppression and optical response.
- [Optical Materials \(2025\)](#): Li-doped CZTS. Bandgap tuned 1.36→1.27 eV via a novel two-step annealing protocol; electrical conductivity  $3.6 \Omega^{-1}\text{cm}^{-1}$ .
- Third first-author article: K-doped CZTS, submitted and currently under review.

#### First-Author, MS Research

- [Ceramics International \(2022\)](#): first report (to the authors' knowledge) of a Sr/Ba/Mn-zirconate perovskite made via coprecipitation instead of the conventional hydrothermal route.

#### Co-Authored & Peer Review

- [Materials Science and Engineering B, 2024](#)
- [Journal of Inorganic and Organometallic Polymers, 2025](#)
- Reviewer, Journal of Materials Science: Materials in Electronics (Springer Nature), 2025.

#### CONFERENCES & AWARDS

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- **2nd Prize, Poster Presentation**, SSCS Conference, COMSATS Abbottabad, 2025.
- **Oral Presentation**, ICRA-C 2024, NUST Islamabad.
- **Poster Presentation**, ICC RTC&RET, University of Swat, 2023.
- Hands-on Training, Advanced Analytical Instrumentation, National Centre for Physics, Islamabad, 2026.